

DSA / Algorithms (Abdul Bari) - Day 7

45-day plan mapped to the Algorithms playlist

Disjoint Sets (Union-Find)

Disjoint set data structure with weighted union and collapsing find.

Watch these Abdul Bari videos

> 1.12 Disjoint Sets Data Structure - Weighted Union and Collapsing Find

Key concepts

- Union-Find tracks a partition of elements into sets.
- Weighted union keeps trees shallow.
- Path compression (collapsing find) flattens trees.

Key idea

`find(x)` with path compression makes future finds near $O(1)$.

Tasks (1.5h)

- ☐ Watch the video
- ☐ Implement union & find
- ☐ Add path compression

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0